17 Scaling up and down: Autoscaling and CloudWatch

This chapter covers

- Creating an Auto Scaling group with a launch template
- Using autoscaling to change the number of virtual machines
- Scaling a synchronous decoupled app behind a load balancer (ALB)
- Scaling an asynchronous decoupled app using a queue (SQS)

Suppose you're organizing a party to celebrate your birthday. How much food and drink do you need to buy? Calculating the right numbers for your shopping list is difficult due to the following factors:

- How many people will attend? You received several confirmations, but some guests will cancel at short notice or show up without letting you know in advance. Therefore, the number of guests is vague.
- How much will your guests eat and drink? Will it be a hot day, with everybody drinking a lot? Will your guests be hungry? You need to guess the demand for food and drink based on experiences from previous parties as well as weather, time of day, and other variables.

Solving the equation is a challenge because there are many unknowns. Being a good host, you'll order more food and drink than needed so no guest will be hungry or thirsty for long. It may cost you more money than necessary, and you may end up wasting some of it, but this possible waste is the risk you must take to ensure you have enough for unexpected guests and circumstances.

Before the cloud, the same was true for our industry when planning the capacity of our IT infrastructure. Planning to meet future demands for your IT infrastructure was nearly impossible. To prevent a supply gap, you needed to add extra capacity on top of the planned demand to prevent running short of resources. When procuring hardware for a data center, we always had to buy hardware based on the demands of the future. We faced the following uncertainties when making these decisions:

- How many users need to be served by the infrastructure?